

CMSC3621 Data Structures and Algorithms Lab

Lab Assignment 3

Overview:

This assignment is for the students to practice back-tracking. The students need to complete the code in two small C++ programs and test them in Visual Studio.

Requirements:

Task 1

Given an array of characters, where every character is unique, please return all possible subsets. The solution set must not contain duplicate subsets. You can return the solution in any order.

For example, if the given array is

[a, b, c]

The resulting vector of vectors should be

[[], [a], [b], [c], [a, b], [a, c], [b, c], [a, b, c]]

You should include two functions in your project:

```
main() //Initialize all variables, invoke the subsets functions,  
      //and output the results.  
vector<vector<char>> subsets(char * array, int size) // Use back-tracking  
                                                    // to get all subsets
```

Task 2

The second task is about the classic algorithm question, Eight Queens Problem. You have learnt how to solve it using back-tracking.

We provide an incomplete implementation of the solution (in EightQueens.zip file). You need to decompress it and load this C++ project into Visual Studio.

To deepen your understanding of back-tracking, we use a format to represent a keyboard, which is different from our textbook. According to the code comments:

The array queens[] is used to record every queen's location. Eg. queens[m]=n means there is a queen on row m, column n.

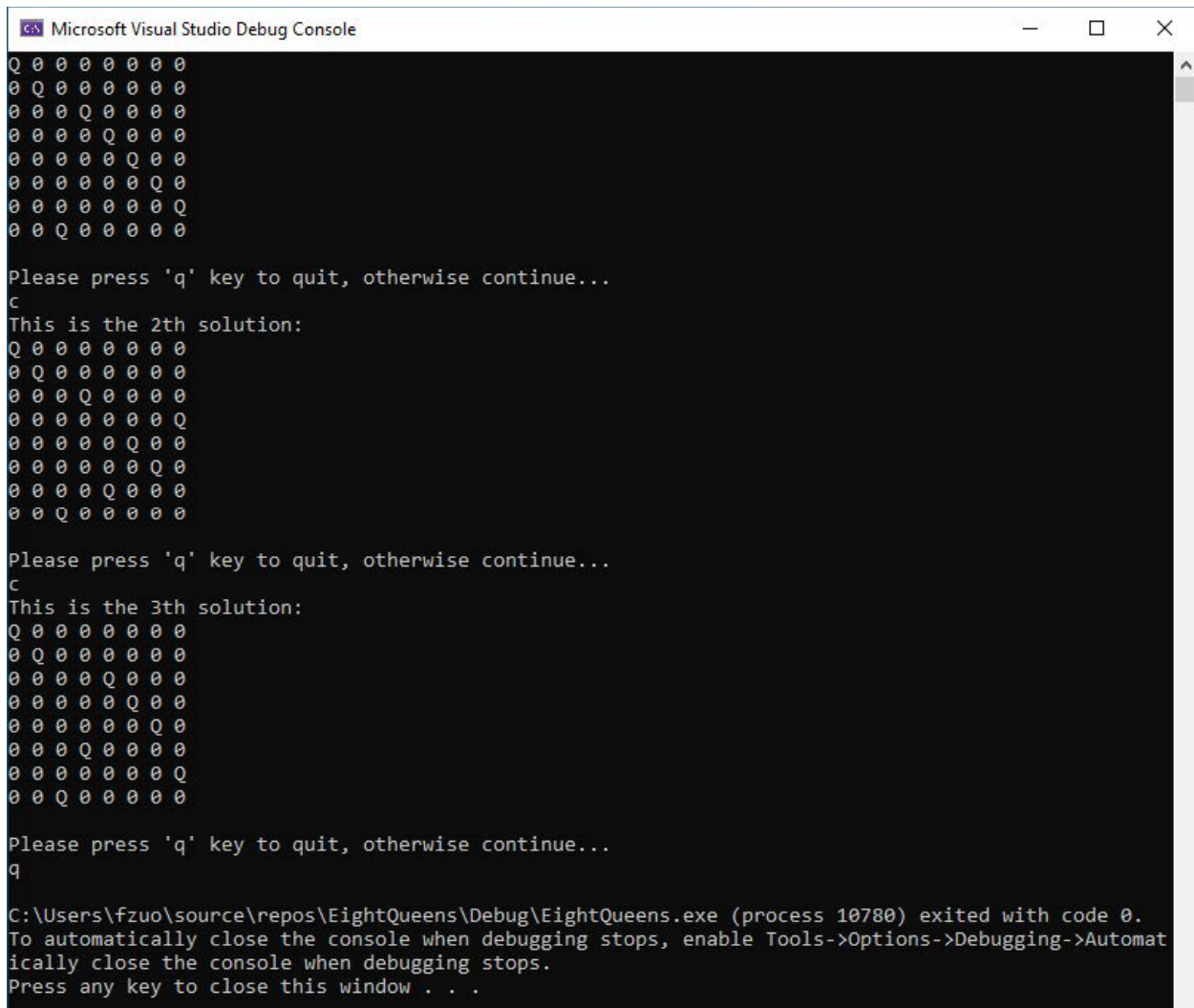
In detail, the QueenPuzzle class has three member functions, where printout() has been provided. You need to complete the other two functions by yourself. You are not allowed to modify any existing parts of the project. You must use comments to explain your code.

Please note that the placeQueen() function should continuously output a new solution until the user press the key 'q' to quit.

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The screenshot below demonstrates the running program. Please note the solutions for Eight Queens in the screenshot may be incorrect. It is just a simulation.



```
Microsoft Visual Studio Debug Console

Q 0 0 0 0 0 0 0 0
0 Q 0 0 0 0 0 0 0
0 0 0 Q 0 0 0 0
0 0 0 0 Q 0 0 0
0 0 0 0 0 Q 0 0
0 0 0 0 0 0 Q 0
0 0 0 0 0 0 0 Q
0 0 Q 0 0 0 0 0 0

Please press 'q' key to quit, otherwise continue...
c
This is the 2th solution:
Q 0 0 0 0 0 0 0 0
0 Q 0 0 0 0 0 0 0
0 0 0 Q 0 0 0 0
0 0 0 0 0 0 0 Q
0 0 0 0 0 Q 0 0
0 0 0 0 Q 0 0 0
0 0 0 0 0 0 Q
0 0 Q 0 0 0 0 0

Please press 'q' key to quit, otherwise continue...
c
This is the 3th solution:
Q 0 0 0 0 0 0 0 0
0 Q 0 0 0 0 0 0 0
0 0 0 0 Q 0 0 0
0 0 0 0 0 Q 0 0
0 0 0 0 0 0 Q
0 0 0 Q 0 0 0 0
0 0 0 0 0 0 Q
0 0 Q 0 0 0 0 0

Please press 'q' key to quit, otherwise continue...
q

C:\Users\fzuo\source\repos\EightQueens\Debug\EightQueens.exe (process 10780) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

Submission:

Submit all three files below to D2L:

1. Archive your Visual Studio project for Task 1 into a zip file. Please use A3_T1.zip as the name.
2. EightQueens.zip file.
3. A report that includes screenshots and explanations to show how you tested your programs.

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Rubrics:

Task1 (35 points); Task2 (50 points); Report (15 points).

Partial credits can be given based on how much you have done.

Any violation against the rules above will lead to extra penalty.

If you missed the report, your submission will not be evaluated.